sDPO: Don't Use Your Data All at Once

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Abstract

As large language models (LLMs) continue to advance, aligning them with human preferences has become a critical objective. In this paper, we introduce stepwise DPO (sDPO), an innovative extension of the recently popularized Direct Preference Optimization (DPO) technique for alignment tuning. sDPO systematically partitions the available preference datasets and applies them incrementally, rather than utilizing the entire dataset simultaneously. This stepwise manner enables the integration of progressively more aligned reference models within the DPO training framework. Our empirical results demonstrate that sDPO not only enhances the alignment precision of reference models but also significantly improves the overall performance of the final model, surpassing other prominent LLMs with larger parameter counts.

1 Introduction

Large language models (LLMs) have revolutionized the field of natural language processing (NLP) by undergoing pre-training, supervised fine-tuning, and alignment tuning, with the latter ensuring the safety and usefulness of the model. Reinforcement learning (RL) techniques (Christiano et al., 2017; Bai et al., 2022), such as proximal policy optimization (PPO) (Schulman et al., 2017), are generally used in this alignment phase.

To address the complicated nature of RL in LLM training, direct preference optimization (DPO) (Rafailov et al., 2023) have been popularized for its simplicity and effectiveness. DPO involves curating preference datasets using human or strong AI (*e.g.,* GPT-4 (OpenAI, 2023)) judgement to select chosen and rejected responses from a pool of multiple answers to a given question. Then, the model being trained (*i.e.*, target model) and a separate reference model compute log probabilities of chosen and rejected responses. Finally, the target model is trained by maximizing the difference of the log probability ratios of the target and the reference models for the chosen and rejected answers. However, obtaining these probabilities can be challenging if one wants to use proprietary models like GPT-4 as the reference model, since they do not offer log probabilities for inputs.

Model	Reference Model	H4
Mistral-7B-OpenOrca	N/A	65.84
Mistral-7B-OpenOrca + DPO	SFT Base	68.87
Mistral-7B-OpenOrca + DPO	SOLAR-0-70B	67.86
Mistral-7B-OpenOrca + DPO	Intel-7B-DPO	**70.13**
OpenHermes-2.5-Mistral-7B	N/A	66.10
OpenHermes-2.5-Mistral-7B + DPO	SFT Base	68.41
OpenHermes-2.5-Mistral-7B + DPO	SOLAR-0-70B	68.90
OpenHermes-2.5-Mistral-7B + DPO	Intel-7B-DPO	**69.72**

Table 1: DPO results in terms of H4 scores for Mistral-7B-OpenOrca and OpenHermes-2.5-Mistral-7B with different reference models. The best results for each SFT base model are shown in bold.

Thus, in practice, the reference model is simply set as the base SFT model (Tunstall et al., 2023; Intel, 2023b; Ivison et al., 2023), which is a much weaker alternative with potentially misaligned preferences. Through Eq. 1, we show that the reference model acts as *a lower bound* in DPO, *i.e.,* the target model is optimized to be at least as aligned as the reference model. Thus, we argue that a reference model that is already more aligned will serve as a better lower bound for DPO training, which would be beneficial for the alignment tuning. One option would be to utilize the plethora of open source models (Tunstall et al., 2023; Ivison et al., 2023) that have already undergone alignment tuning.

Note that the above approach may not be feasible due to the absence of such aligned models, or the fact that it renounces control over the reference model, which could lead to safety concerns. Instead, we propose 'stepwise DPO', named sDPO, where we use the preference datasets (or subsets

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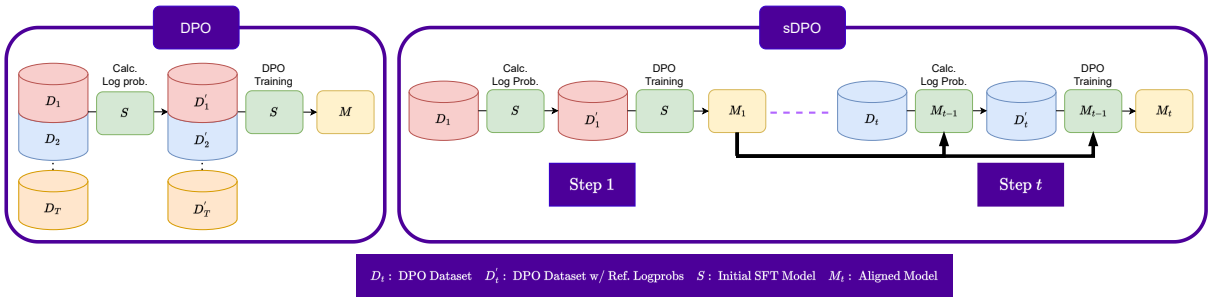

Figure 1: Overview of sDPO where preference datasets are divided to be used in multiple steps. The aligned model from the previous step is used as the reference and target models for the current step. The reference model is used to calculate the log probabilities and the target model is trained using the preference loss of DPO at each step.

of a preference dataset) in *a step-by-step manner* rather than all at once when undergoing DPO training. The aligned model in the previous step is used as the reference model for the current step, which results in utilizing a more aligned reference model (*i.e.*, a better lower bound). Empirically, we show that using sDPO results in a more performant final aligned model as well.

While concurrent works (Yuan et al., 2024) that focus on an iterative pipeline of generating *new* preference data have been proposed, our method focuses on utilizing the *currently available* preference datasets. Thus, our approach is complementary as sDPO can be easily applied to any preference data and further combination with concurrent works would be an exciting future direction.

2 Related Work

2.1 Large Language Models

Recent research has highlighted a "scaling law" in the field of context-based language models (Kaplan et al., 2020; Hernandez et al., 2021; Anil et al., 2023), showing a proportional relationship between the size of the model plus the training data and the resulting performance improvements. Consequently, this has led to the advent of LLMs. In contrast to earlier models, LLMs can perform in-context learning, which includes abilities such as zero-shot learning (Radford et al., 2019) and few-shot learning (Brown et al., 2020), allowing them to adapt and perform tasks without the need for weight adjustments. These emergent abilities of LLMs, absent in their smaller counterparts, signal a significant evolution in language model capabilities (Wei et al., 2022).

2.2 Alignment Tuning

LLMs have been recognized to produce text that may seem linguistically inconsistent to human interpreters because their pretraining is based not on an understanding of human intentions but on a broad spectrum of domain-specific knowledge, as indicated in (Ziegler et al., 2019). In an effort to rectify this issue and better mirror human intentions, prior research (Ziegler et al., 2019) has suggested the adoption of Reinforcement Learning with Human Feedback (RLHF). RLHF seeks to refine the LLM's output by constructing a reward model that aligns with human preferences and applying reinforcement learning to direct the LLM towards selections that garner the most favorable reward metrics. This approach is intended to bolster the safety, decorum, and general excellence of the responses produced by the LLM. Nonetheless, despite showing promising results, RLHF is confronted with challenges, such as the intricate handling of an extensive set of hyperparameters and the necessity to amalgamate several models (policy, value, reward, and reference models).

To address these issues, there have been proposals for supervised fine-tuning methodologies such as RRHF (Yuan et al., 2023), RAFT (Dong et al., 2023), and DPO (Rafailov et al., 2023). These methods circumvent the intricacies inherent in reinforcement learning and have been shown to yield empirical results on par with RLHF. Notably, the DPO technique straightforwardly encourages the LLM to favor positive responses and discourage negative ones. DPO has been observed to yield performant learning outcomes, in spite of its uncomplicated training procedure.

Concurrent to our work, Yuan et al. (2024) have developed an iterative framework for generating *new* preference datasets and performing

DPO training on the resulting datasets. They empirically demonstrated the superiority of their iterative framework in terms of AlpacaEval 2.0. In contrast, our work is complementary to the above in the sense that we focus on utilizing the *current* preference data and does not undergo new data generation. Thus, our method can also be applied to Yuan et al. (2024) by changing the DPO training part to using sDPO instead. Additionally, the evaluation used in Yuan et al. (2024) is also different to ours as we utilize tasks from Open LLM Leaderboard (Beeching et al., 2023), EQ Bench (Paech, 2023) and MT Bench (Zheng et al., 2023) whereas Yuan et al. (2024) uses AlpacaEval 2.0.

3 Methodology

3.1 Preliminary Investigation on Reference Models

To gauge the importance of using a well-aligned reference model in DPO, we perform preliminary experiments of DPO training with the Ultrafeedback dataset (Cui et al., 2023) on Mistral-7B-OpenOrca (Lian et al., 2023) and OpenHermes-2.5-Mistral-7B (Teknium, 2023) as the SFT base model, owing to their excellent performance and small size. We compare the following reference models: i) the SFT base model itself, same as the conventional DPO setup; ii) SOLAR-0-70B (Upstage, 2023), a larger and much more performant model; and iii) Intel-7B-DPO (Intel, 2023a), an already aligned reference model. The results are summarized in Table 1.

As the table shows, using Intel-7B-DPO as the reference model results in the best performance, even better than using SOLAR-0-70B, which is a much larger and performant model. Thus, whether the reference model is pre-aligned or not plays an important role in the resulting aligned model's performance. Unfortunately, it is not always possible to use a open sourced pre-aligned model as the reference model due to technical and safety concerns. For instance, such a model may not exist yet or can be susceptible to various domain-specific harmfulness and fairness criteria along with potential data contamination issues. To circumvent the above, we propose sDPO, which does not require an external pre-aligned model but uses more aligned reference models, built from the SFT base model, as a part of the training framework.

3.2 Stepwise DPO

In sDPO, we propose to use the available preference datasets in a stepwise manner instead of using them all at once. Essentially, we partition the preference data into T chunks and perform DPO training T times. The trained model from the previous step is used as the reference and target models, which means that each of the T DPO training steps function in a similar manner to the conventional DPO setup. In doing so, we create and utilize intermediary reference models that are more aligned than those thar are used in conventional DPO. The comparison of the overall flow of DPO and sDPO is presented in Figure 1.

Reference model. The reference model is used to calculate the log probabilities of the preference dataset. For each step, only a subset of the total data is used and the reference model is initialized as M_{t-1}, *i.e,* the aligned model from the previous step. The initial reference model is set as S, the SFT base model. This results in using a more aligned reference model than conventional DPO.

Target model. For $t > 1$, the target model which is trained using the preference loss of DPO in each step of sDPO is also initialized as M_{t-1} instead of S. This ensures that the final model trained with sDPO has been directly trained with the same amount data as a model trained with DPO.

Mathematical explanation. To gain a deeper understanding of sDPO, we rearrange the DPO loss from (Rafailov et al., 2023), as follows:

$$\begin{aligned}
&\mathcal{L}_{\text{DPO}}(\pi_\theta, \pi_{ref}) \\
&= -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w|x)}{\pi_{ref}(y_w|x)} - \beta\log\frac{\pi_\theta(y_l|x)}{\pi_{ref}(y_l|x)}\right)\right] \\
&= -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\cdot(\gamma_{\pi_\theta}(x,y_w,y_l) - \gamma_{\pi_{ref}}(x,y_w,y_l))\right)\right],
\end{aligned} \tag{1}$$

where D is the preference dataset, x is the question, y_w and y_l are the chosen and rejected answers respectively, θ is the learnable parameters of the model, and $\gamma_\pi(x, y_w, y_l) = \log\frac{\pi(y_w|x)}{\pi(y_l|x)}$, *i.e.,* the logratio of the chosen and rejected samples w.r.t. the policy π. As $\log\sigma(\cdot)$ is a monotonically increasing function and $\gamma_{\pi_{ref}}$ is fixed before training, the minimization of $\mathcal{L}_{\text{DPO}}(\pi_\theta, \pi_{ref})$ leads to $\gamma_{\pi_\theta} > \gamma_{\pi_{ref}}$ on average. Thus, $\gamma_{\pi_{ref}}$ can be understood as a lower bound defined by the reference model, of which the target model is trained such that $\gamma_{\pi_\theta} > \gamma_{\pi_{ref}}$. In sDPO, $\gamma_{\pi_{ref}}$ increases as the steps progress because the reference model that defines it is more and more aligned. Hence, $\gamma_{\pi_{ref}}$ becomes a stricter lower bound as the steps pass,

Model	Size	Type	H4 (Avg.)	ARC	HellaSwag	MMLU	TruthfulQA
SOLAR 10.7B + SFT + sDPO	$\sim$ 11B	Alignment-tuned	**74.31**	**71.33**	88.08	65.39	**72.45**
SOLAR 10.7B + SFT + DPO	$\sim$ 11B	Alignment-tuned	72.67	69.62	87.16	66.00	67.90
Mixtral 8x7B-Instruct-v0.1	$\sim$ 47B	Alignment-tuned	73.40	70.22	87.63	71.16	64.58
SOLAR-0-70B-16bit	$\sim$ 70B	Instruction-tuned	72.93	71.08	87.89	70.58	62.25
Qwen 72B	$\sim$ 72B	Pretrained	72.17	65.19	85.94	**77.37**	60.19
Yi 34B	$\sim$ 34B	Pretrained	70.72	64.59	85.69	76.35	56.23
SOLAR 10.7B + SFT	$\sim$ 11B	Instruction-tuned	69.51	67.32	85.96	65.95	58.80
Mistral 7B-Instruct-v0.2	$\sim$ 7B	Instruction-tuned	69.27	63.14	84.88	60.78	68.26
Falcon 180B	$\sim$ 180B	Pretrained	68.57	69.45	**88.86**	70.50	45.47
Mixtral 8x7B-v0.1	$\sim$ 47B	Pretrained	67.78	66.04	86.49	71.82	46.78
Llama 2 70B	$\sim$ 70B	Pretrained	67.35	67.32	87.33	69.83	44.92
Zephyr	$\sim$ 7B	Alignment-tuned	66.36	62.03	84.52	61.44	57.44
Qwen 14B	$\sim$ 14B	Pretrained	64.85	58.28	83.99	67.70	49.43
SOLAR 10.7B	$\sim$ 11B	Pretrained	64.27	61.95	84.60	65.48	45.04
Mistral 7B	$\sim$ 7B	Pretrained	62.40	59.98	83.31	64.16	42.15

Table 2: Performance comparison of applying sDPO or DPO to SOLAR 10.7B + SFT against various top performing models. Size is shown in units of billions of parameters and type is reported as one of {'Pretrained', 'Instruction-tuned', 'Alignment-tuned'}. Models based on SOLAR 10.7B are shown in purple color. The best scores in each column are shown in bold.

inducing a *curriculum learning* from easy to hard optimization tasks. Thus, the target model is being trained to learn stricter preferences as the steps progress in sDPO.

Data partitioning strategy. The method for partitioning the preference data into T chunks is also important in sDPO. One option would be to pool all the data from different dataset sources and perform random sampling. However, we argue that partitioning the data such that earlier chunks are comprised of easier preference data would be more aligned with inducing a curriculum learning of easy to hard optimization in sDPO.

To that end, we propose to use easy-to-hard data partitioning by the following method. Using M_0, the initial target model, we calculate the reward accuracy, *i.e.,* the percentage of samples in which the target model scores higher rewards for preferred samples, for the different dataset sources. The dataset sources are sorted in descending order of the reward accuracy, which are then used as the T chunks in sDPO. Thus, if we have N dataset sources, we would have a total of N chunks, where earlier chunks would contain easier samples as measured by the reward accuracy.

4 Experiments

4.1 Experimental Setup

Training details. We use a supervised fine-tuned SOLAR 10.7B (Kim et al., 2023) as our SFT base model S as it delivers excellent performance with its uncommon yet relatively small 10.7B size. Note that we do not need a separate reference model as it is initialized as M_{t-1}, the final trained model from the previous step. We use OpenOrca (Mukherjee et al., 2023) ($\sim 12K$ samples) and Ultrafeedback Cleaned ($\sim 60K$ samples) (Cui et al., 2023; Ivison et al., 2023) as our preference datasets. The training hyper-parameters follow that of Tunstall et al. (2023). Using the easy-to-hard partitioning, we use OpenOrca as dataset D_1 and Ultrafeedback Cleaned as dataset D_2.

Evaluation. We mainly utilize four log-probability tasks in the HuggingFace Open LLM Leaderboard (Beeching et al., 2023): ARC (Clark et al., 2018), HellaSWAG (Zellers et al., 2019), MMLU (Hendrycks et al., 2020), TruthfulQA (Lin et al., 2022). We also report the average scores for the four tasks, which is denoted as H4. Note that these tasks do not require the model to actually generate a new answer to the question. Rather, the log-probability of a pre-defined answer is measured instead.

To augment the above potential downside of log-probability benchmarks, we also incorporate generation benchmarks such as EQ Bench (Paech, 2023) and MT Bench (Zheng et al., 2023), where a model is prompted to generate an answer to a question. As such, MT Bench and EQ Bench both strongly correlate with the Chatbot Arena ELO (Zheng et al., 2023; Chiang et al., 2024), one of the most widely recognized open-world LLM evaluation system.

4.2 Main Results

Evaluation results for applying sDPO to the SFT base model, along with results for other top-

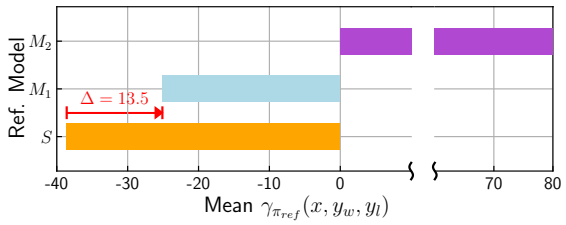

Figure 2: Mean $\gamma_{\pi_{ref}}$ on Ultrafeedback Cleaned dataset for different reference models S, M_1, and M_2. Note that the x-axis is in log scale.

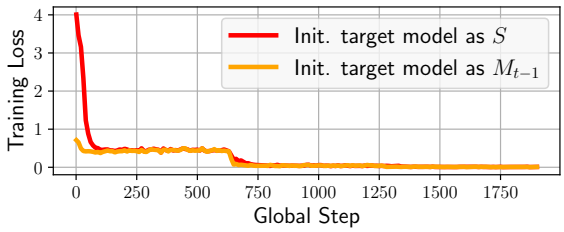

Figure 3: Loss curve comparison in step 2 of sDPO for different initializations of the target model.

performing models are shown in Table 2. Applying sDPO on SOLAR 10.7B + SFT further increases the H4 score to 74.31, an improvement of +4.80. Notably, 'SOLAR 10.7B + SFT + sDPO' outperforms other larger models such as Mixtral 8x7B-Instruct-v0.1, despite the smaller number of parameters. This highlights that effective alignment tuning could be the key to unlocking next level performance for smaller LLMs. Further, applying sDPO results in substantially higher score of 72.45 for TruthfulQA, which demonstrates the effectiveness of the alignment tuning process. We also present additional results in Table 4 of Section 4.7 on the EQ Bench (Paech, 2023), which is a generation task with high correlation with the Chatbot Arena ELO (Zheng et al., 2023). The additional results indicate the superiority of sDPO over DPO in improving generation task performance as well.

4.3 Ablation Studies Against DPO

We also report evaluation results for ablating sDPO with traditional DPO in Table 2. 'SOLAR 10.7B + SFT + DPO' uses all the DPO data at once, *i.e.,* $D_1 + D_2$, same as the conventional DPO training setup.

We can see that using sDPO over DPO results in a higher H4 score overall, with noticeable improvements in ARC and TruthfulQA scores. Therefore, we believe sDPO could function as a drop-in replacement for DPO training with better performance.

4.4 Reference Models in sDPO

Effectiveness of sDPO in terms of alignment tuning. In Sec. 3.2, we explain that the reference models in sDPO are more aligned, resulting in higher $\gamma_{\pi_{ref}}$, *i.e.,* a stricter lower bound. We verify the above empirically in Figure 2 by comparing the mean $\gamma_{\pi_{ref}}$ on the Ultrafeedback Cleaned dataset for the reference models in steps 1 and 2 of sDPO, *i.e.,* S and M_1. Note that these two models have not been trained on the aforementioned dataset. Using the SFT base model S as the reference model, the mean of $\gamma_{\pi_{ref}}$ is -38.60. On the other hand, using the aligned model M_1 from step 1 of sDPO as the reference model, the mean of $\gamma_{\pi_{ref}}$ is -25.10, an increase of 13.50 in *log scale*. Thus, a single step of sDPO greatly increases $\gamma_{\pi_{ref}}$, which results in a more performant aligned model as seen in Table 2.

Adopting open source models as reference models could be dangerous. We also show mean $\gamma_{\pi_{ref}}$ of M_2, the aligned model from step 2 of sDPO. Unlike S and M_1, M_2 is trained on the Ultrafeedback Cleaned dataset, *i.e.,* M_2 is used as a reference model on data that was *already used to train it*. Note that such a case could happen commonly when adopting various open source models as reference models. This is because the datasets that were used in training those models are often unclear and could overlap with the preference datasets unintentionally. Mean $\gamma_{\pi_{ref}}$ of M_2 is 84.35, which is staggeringly higher than either S or M_1. The strikingly high value for M_2 likely points to overfitting of M_2 to the Ultrafeedback Cleaned dataset. Note that utilizing such an absurdly high value of $\gamma_{\pi_{ref}}$ as the lower bound in DPO training may be undesirable. This result highlights the potential danger of merely adopting open source models as reference models instead of using sDPO.

4.5 Target Model Initialization in sDPO

One option for target model initialization in sDPO is to use S, the initial SFT base model, for *all steps*. However, such initialization results in the final model trained with sDPO seeing less data than using DPO instead. Further, the target model and the reference model become more and more different as the steps progress, which is a deviation from the original DPO setup and risks losing the theoretical benefits of DPO.

Model	H4 (Avg.)	ARC	HellaSwag	MMLU	TruthfulQA
SOLAR 10.7B + SFT + sDPO	**74.31**	**71.33**	**88.08**	65.39	**72.45**
SOLAR 10.7B + SFT + sDPO Rand.	72.56	69.20	87.27	**65.96**	67.81

Table 3: Comparison between the easy-to-hard and random partitioning strategies. 'SOLAR 10.7B + SFT + sDPO' uses the easy-to-hard partitioning whereas 'SOLAR 10.7B + SFT + sDPO Rand.' denotes sDPO with random partitioning instead. Easy-to-hard partitioning is better than random partitioning. The best scores are shown in bold.

Model	EQ Bench	MT Bench
SOLAR 10.7B + SFT + sDPO	**68.83**	**7.43**
SOLAR 10.7B + SFT + DPO	61.02	7.35
SOLAR 10.7B + SFT	60.48	7.14

Table 4: Additional results on EQ Bench (Paech, 2023) and MT Bench (Zheng et al., 2023), both of which are generation tasks that highly correlate with the Chatbot Arena ELO (Zheng et al., 2023; Chiang et al., 2024). The best scores for both benchmarks are shown in bold.

To concretely investigate such potential issues, we visualize the loss curves for initializing the target model as S in Figure 3. We observe that the initial loss value is much higher when compared to initializing the target model as M_{t-1}, *i.e.,* the same as the reference model and adhering to the DPO convention. As using M_{t-1} the target model means that each *step* of sDPO is using the same setup as DPO, the loss curves are much more stable and desirable. Thus, for stable training, initializing the target model as M_{t-1} was chosen for sDPO.

4.6 Easy-to-Hard Data Partitioning

The effectiveness of the easy-to-hard data partitioning used in sDPO is demonstrated in Table 3. Note that we use OpenOrca as D_1 and Ultrafeedback Cleaned as D_2. As 'SOLAR 10.7B + SFT + sDPO', which uses the easy-to-hard partitioning, is more performant than'SOLAR 10.7B + SFT + sDPO Rand.', which uses random partitioning, the proposed easy-to-hard data partitioning is more effective for sDPO.

4.7 Additional Results on Generation Tasks

In Table 4, we also report results for EQ Bench (Paech, 2023) and MT Bench (Zheng et al., 2023) for the SFT base model and the models obtained by applying DPO and sDPO on the SFT base model.

For EQ Bench, we use the version without the revision prompt. We note that the EQ Bench requires the models to generate an answer that can be parsed with a pre-defined template for evaluation, which could be said to measure distinct capabilities of LLMs from the log-probability benchmarks shown in Table 2. While applying DPO only mildly improves the performance from the SFT base model, applying sDPO improves the performance significantly by over $+8\%$, indicating the effectivenss in which sDPO improves the generation capabilities compared to DPO.

As for MT Bench, we note that using sDPO achieves the best score of 7.43 amongst the compared models. Notably, applying sDPO to the SFT base model improves the MT Bench score by a non-trivial margin of $+0.29$. Applying DPO to the SFT base model also improves the MT Bench score, but not by more than that of applying sDPO.

5 Conclusion

We propose sDPO, an extension of DPO for aligning LLMs. Unlike traditional DPO, sDPO employs a stepwise approach, using subsets of preference data sequentially. This method leverages the aligned model from the previous step as the reference for the current step, ensuring progressively better alignment. Our experiments demonstrate that sDPO significantly outperforms conventional DPO in terms of both log-probability benchmarks such as ARC, HellaSWAG, MMLU, and TruthfulQA, as well as generation benchmarks such as EQ Bench and MT Bench. Additionally, sDPO enhances model alignment, as indicated by higher mean $\gamma_{\pi_{ref}}$ values, showing improved alignment with human preferences. The stepwise nature of sDPO simplifies the training process and aligns with curriculum learning principles, facilitating a structured optimization path. By using existing preference datasets more effectively, sDPO results in higher performance and better-aligned language models. This approach has the potential to transform alignment tuning, offering a robust framework for future research in LLMs.

Limitations

While we have demonstrated the effectiveness of employing easy-to-hard data partitioning of different datasets in distinct stages of sDPO, identifying a more performant strategy for segmenting more intricate preference data collections remains an area for further exploration.

Furthermore, our experiments predominantly utilized SOLAR 10.7B models, driven by the state-of-the-art performance at the time of experimentation along with its relatively 10.7 billion parameter size. Although as SOLAR 10.7B models are also based on the Llama-2 architecture with our results likely to transfer to other similar decoder only transformer models, more experiments using other models would be beneficial.

Additionally, as with most research on LLMs, we operated within our limitations in computational resources. Although this focus has yielded significant insights, expanding our experimental framework to incorporate a broader range of Large Language Models (LLMs) could potentially unveil more comprehensive understanding of the strengths and limitations of sDPO. Such an expansion would allow for a more robust comparison across different model architectures and sizes, further enriching our findings.

Evaluating the efficacy of LLMs is an evolving challenge in the field. In our study, we primarily employed tasks from the Huggingface Open LLM Leaderboard as benchmarks for evaluation along with EQ Bench and MT Bench. While this provided comparative results, future research could benefit from incorporating a wider array of tasks and benchmarks. These could include tasks that judge actual human or strong AI preference alignment. Such additional evaluation would not only enhance the validity of our findings but also contribute to the broader discourse on LLM assessment methodologies.

Ethics Statement

In this study, we strictly adhered to ethical standards in the conduct of our research. Our experiments were based entirely on open models and open datasets, ensuring transparency and accessibility. We took meticulous care to avoid any biases or data contamination, thereby maintaining the integrity of our research process. The experimental environment was rigorously designed to be objective, ensuring that all comparisons conducted were fair and impartial. This approach reinforces the reliability and validity of our findings, contributing positively to the field while upholding the highest ethical standards. We confirmed that all the data used in our experiments were free of licensing issues.

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